



Mathematics 2 (5EZB0)

Studio classroom 2 – Week 6

May 28, 2026

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Topics for today

- Vector functions of one variable and parametrizations (Adams 12.1 & 12.3)
- Gradient and directional derivatives (Adams 13.7 + extra)

Vector functions of one variable and parametrization

Unfinished business from Studio classroom week 2

Studio classroom – Problem 1

The position, velocity, and acceleration of a particle moving in 3D are given by $\mathbf{r}(t)$, $\mathbf{v}(t)$, and $\mathbf{a}(t)$ as a function of time t , respectively. Furthermore, it is given that at every time t , the acceleration $\mathbf{a}$ is perpendicular to both $\mathbf{r}$ and $\mathbf{v}$.

(a) Show that the vector $\mathbf{r}(t) - t\mathbf{v}(t)$ has constant length.

Studio classroom – Problem 2

The position of an object as a function of time t can be described by the piecewise smooth curve $\mathbf{r} = t^3\hat{\mathbf{i}} + t^2\hat{\mathbf{j}}$.

- (a) Give expressions for the velocity $\mathbf{v}$ and acceleration $\mathbf{a}$ of the object as a function of time.
- (b) Calculate the distance the object has travelled (the length of the curve) between $t = -1$ and $t = 2$ s.

Gradient

Cartesian coordinates

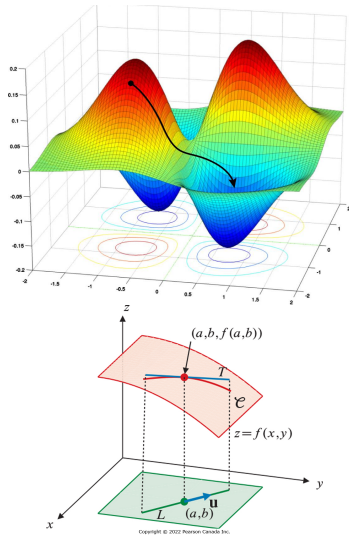
Gradient in Cartesian coordinates

Gradient operator ∇ (or grad):

$$\nabla f = \text{grad } f = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) f$$

Properties:

- operates on **scalar field** $f = f(x, y, z)$ to yield a **vector field**
- direction of ∇f :
 - direction in which f increases most rapidly
 - perpendicular to level sets



Gradient: example in 2D

For $0 < x < 1$ and $0 < y < 2$, consider the function

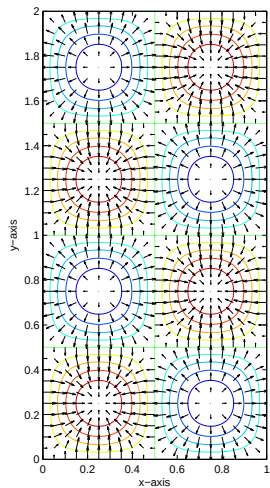
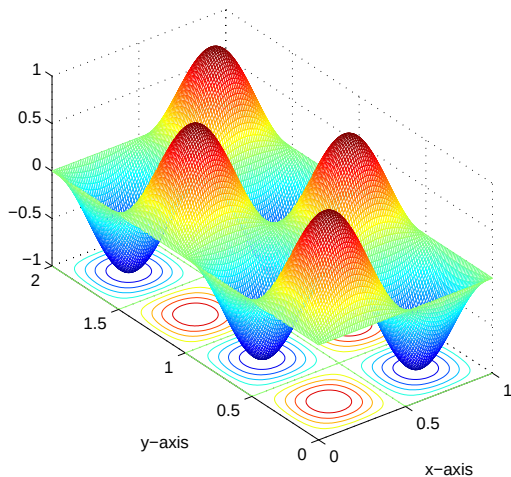
$$\Psi(x,y) = \sin(2\pi x) \sin(2\pi y)$$

Use MATLAB to make plots of

- contours $\Psi(x,y) = \Psi_0$ ("level map")
- the gradient

$$\nabla \Psi(x,y) = 2\pi \left[\cos(2\pi x) \sin(2\pi y) \hat{i} + \sin(2\pi x) \cos(2\pi y) \hat{j} \right]$$

Gradient: example in 2D

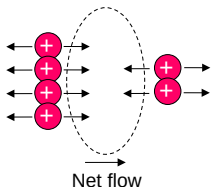


Gradients in physics

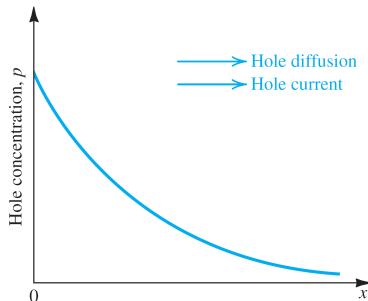
Examples

Diffusion current (semiconductor physics)

- ▶ Diffusion is based on **concentration difference**, not on charge of carriers
- ▶ Temperature will cause *all* carriers to move in *all* directions



- ▶ Local concentration difference will lead to net flow of carriers in direction of lower concentration



Sedra & Smith, Fig 1.33

Diffusion current, equations

Diffusion constant (diffusivity): $D = \frac{kT}{q}\mu$ unit: cm^2/s

$$n_{\text{diffusion}} \stackrel{\text{def}}{=} -D_n \frac{dn}{dx} \quad n, p = \text{carriers / volume}$$

$$p_{\text{diffusion}} \stackrel{\text{def}}{=} -D_p \frac{dp}{dx} \quad n, p_{\text{diffusion}} = \text{diffusive flux}$$

$$J_{n,\text{diffusion}} = qD_n \frac{dn}{dx} = kT\mu_n \frac{dn}{dx}$$

$$J_{p,\text{diffusion}} = -qD_p \frac{dp}{dx} = kT\mu_p \frac{dp}{dx}$$

$$J_{\text{diffusion}} = kT \left(\mu_n \frac{dn}{dx} - \mu_p \frac{dp}{dx} \right)$$

(Find current I by integrating current density J over device area)

Diffusion: Fick's law

$\varphi(x, t)$: concentration of particles

$J_x(x, t)$: **diffusive flux** in x -direction

(number of particles moving per unit of area per unit of time)

Fick's law in one dimension: $J_x(x, t) = -D \frac{d\varphi(x, t)}{dx}$

with D the diffusivity

Fick's law generalized to three dimensions: $\mathbf{J} = -D \nabla \varphi$

$-\nabla \varphi$: local direction towards lower concentration

Conservative forces (one dimension)

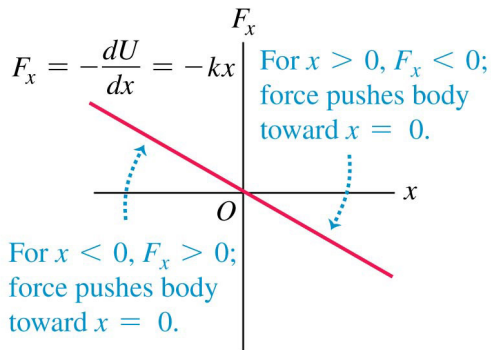
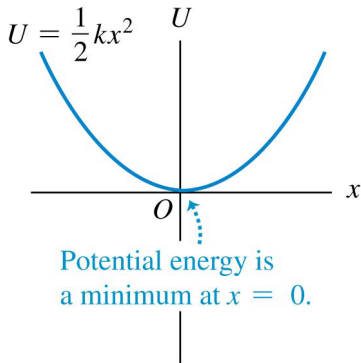
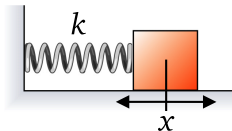
- ▶ **Conservative force** at position x can be obtained from potential energy function as function of x

$$F_x(x) = -\frac{dU(x)}{dx}$$

- ▶ In regions where $U(x)$ changes rapidly with $x \Rightarrow$ large force (magnitude)
- ▶ When $F_x(x)$ is in positive x -direction, $U(x)$ decreases with increasing x
- ▶ Conservative force always acts to push the system **towards lower potential energy**

One dimensional: elastic potential energy

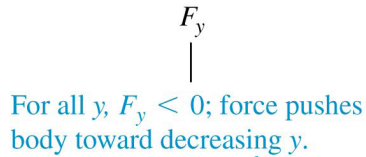
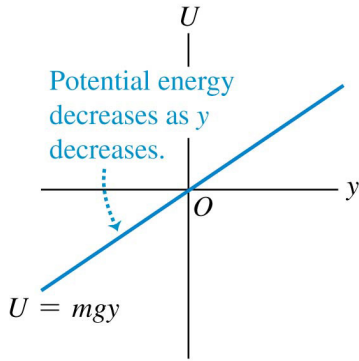
$$F = F_x = -\frac{dU}{dx}$$



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One dimensional: gravitational potential energy

$$F = F_y = -\frac{dU}{dy}$$



$$F_y = -\frac{dU}{dy} = -mg$$

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General approach: three dimensions

Potential energy function can depend on 3D position (e.g. Cartesian) as $U = U(x, y, z)$

The three force components of force vector $\mathbf{F}$ at any point (x, y, z) in space then are

$$F_x = -\frac{\partial U}{\partial x}, \quad F_y = -\frac{\partial U}{\partial y}, \quad F_z = -\frac{\partial U}{\partial z}$$

$$\mathbf{F} = -\left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)U = -\nabla U$$

Studio classroom – Problem 3

A static electric field $\mathbf{E}$ is also a conservative field, which can be determined from a known electric potential V using $\mathbf{E} = -\nabla V$

Calculate the electric field in Cartesian coordinates for the following two cases:

- (a) An infinitely long line carrying a constant charge density ρ_L (C/m), for which the electric potential is given by $V = -\frac{\rho_L}{2\pi\epsilon_0} \ln \sqrt{x^2 + y^2}$.
- (b) A point charge q (C) in the origin, for which the electric potential is given by $V = \frac{q}{4\pi\epsilon_0 \sqrt{x^2 + y^2 + z^2}}$

Gradient

Cylindrical coordinates

Cylindrical coordinates (recap)

Coordinates

$$x = r \cos \theta$$

$$y = r \sin \theta$$

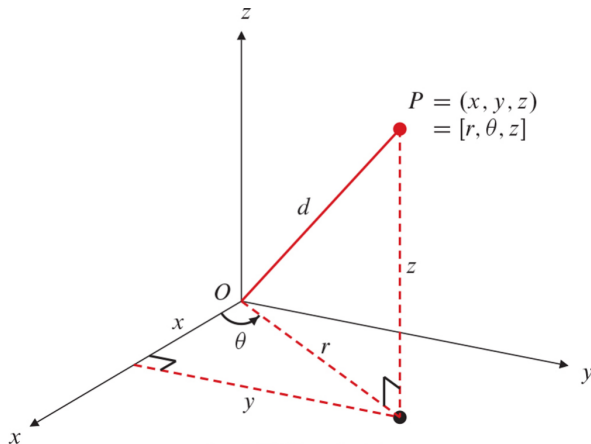
$$z = z$$

Unit vectors

$$\hat{\mathbf{r}} = \cos \theta \hat{\mathbf{i}} + \sin \theta \hat{\mathbf{j}}$$

$$\hat{\boldsymbol{\theta}} = -\sin \theta \hat{\mathbf{i}} + \cos \theta \hat{\mathbf{j}}$$

$$\hat{\mathbf{k}} = \hat{\mathbf{k}}$$



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Gradient in cylindrical coordinates

Two approaches possible:

1. Following formal mathematical derivation (see Math 2 lectures) $\Rightarrow$ brief summary included
2. Easier, faster derivation (more intuitive, hopefully) $\Rightarrow$ detailed derivation

Similar approaches could be used for spherical coordinates

Approach 1 – Gradient in cylindrical coordinates

Start from the gradient operator in Cartesian coordinates

$$\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

Express this in terms of cylindrical coordinates, for which the gradient operator needs to look like

$$\nabla = a \hat{r} \frac{\partial}{\partial r} + b \hat{\theta} \frac{\partial}{\partial \theta} + c \hat{k} \frac{\partial}{\partial z}$$

and determine ‘scaling’ factors a, b, c

Approach 1 – Gradient in cylindrical coordinates (cont'd)

Use chain rule for derivatives

$$\frac{\partial}{\partial r} = \frac{\partial x}{\partial r} \frac{\partial}{\partial x} + \frac{\partial y}{\partial r} \frac{\partial}{\partial y} + \frac{\partial z}{\partial r} \frac{\partial}{\partial z}, \quad \frac{\partial}{\partial \theta} = \frac{\partial x}{\partial \theta} \frac{\partial}{\partial x} + \frac{\partial y}{\partial \theta} \frac{\partial}{\partial y} + \frac{\partial z}{\partial \theta} \frac{\partial}{\partial z}$$
$$\frac{\partial}{\partial z} = \frac{\partial}{\partial z}$$

and *several square metres of blackboard* to arrive at

$$\nabla = \hat{r} \frac{\partial}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{k} \frac{\partial}{\partial z}$$

Approach 2 – Intermezzo: Cartesian example

Consider scalar function of three variables $f(x,y,z)$

- Δf in going from (x,y,z) to $(x + \Delta x, y + \Delta y, z + \Delta z)$:

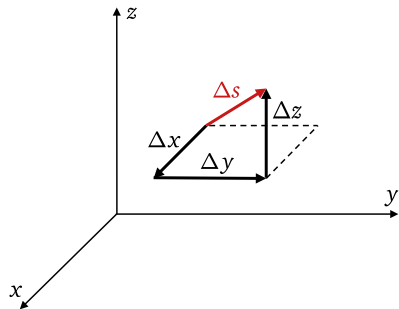
$$\Delta f = \frac{\partial f}{\partial x} \Delta x + \frac{\partial f}{\partial y} \Delta y + \frac{\partial f}{\partial z} \Delta z + \dots$$

- Vector displacement

$$\Delta \mathbf{s} = \hat{\mathbf{i}} \Delta x + \hat{\mathbf{j}} \Delta y + \hat{\mathbf{k}} \Delta z$$

allows us to write

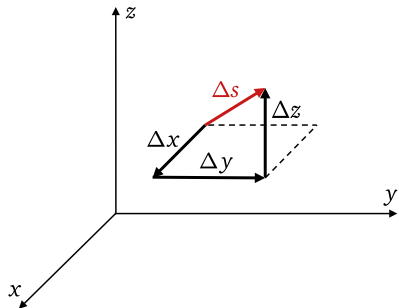
$$\Delta f = \left(\hat{\mathbf{i}} \frac{\partial f}{\partial x} + \hat{\mathbf{j}} \frac{\partial f}{\partial y} + \hat{\mathbf{k}} \frac{\partial f}{\partial z} \right) \cdot \Delta \mathbf{s} + \dots$$



Approach 2 – Intermezzo: Cartesian example (cont'd)

- Next, write $\Delta \mathbf{s} = \hat{\mathbf{u}} \Delta s$ (unit vector times length) and take the limit

$$\lim_{\Delta s \rightarrow 0} \frac{\Delta f}{\Delta s} = \frac{df}{ds} = \underbrace{\left(\hat{\mathbf{i}} \frac{\partial f}{\partial x} + \hat{\mathbf{j}} \frac{\partial f}{\partial y} + \hat{\mathbf{k}} \frac{\partial f}{\partial z} \right)}_{\text{gradient}} \cdot \hat{\mathbf{u}}$$



Approach 2 – Gradient in cylindrical coordinates

Consider scalar function of three variables $f(r, \theta, z)$

- Δf in going from (r, θ, z) to $(r + \Delta r, \theta + \Delta \theta, z + \Delta z)$:

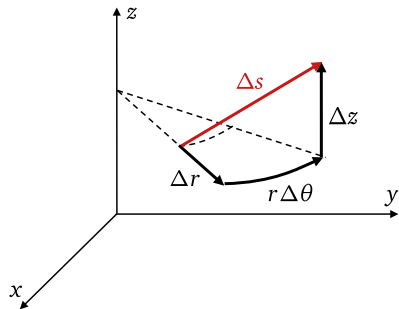
$$\Delta f = \frac{\partial f}{\partial r} \Delta r + \frac{\partial f}{\partial \theta} \Delta \theta + \frac{\partial f}{\partial z} \Delta z + \dots$$

- Vector displacement

$$\Delta \mathbf{s} = \hat{\mathbf{r}} \Delta r + \hat{\boldsymbol{\theta}} r \Delta \theta + \hat{\mathbf{k}} \Delta z$$

allows us to write

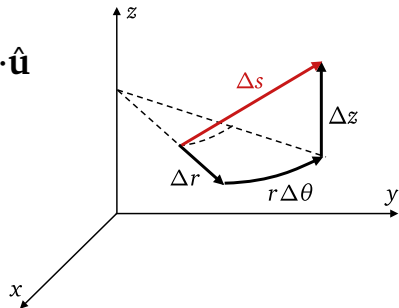
$$\Delta f = \left(\hat{\mathbf{r}} \frac{\partial f}{\partial r} + \hat{\boldsymbol{\theta}} \frac{1}{r} \frac{\partial f}{\partial \theta} + \hat{\mathbf{k}} \frac{\partial f}{\partial z} \right) \cdot \Delta \mathbf{s} + \dots$$



Approach 2 – Gradient in cylindrical coordinates (cont'd)

- Next, write $\Delta \mathbf{s} = \hat{\mathbf{u}} \Delta s$ (unit vector times length) and take the limit

$$\lim_{\Delta s \rightarrow 0} \frac{\Delta f}{\Delta s} = \frac{df}{ds} = \underbrace{\left(\hat{\mathbf{r}} \frac{\partial f}{\partial r} + \hat{\boldsymbol{\theta}} \frac{1}{r} \frac{\partial f}{\partial \theta} + \hat{\mathbf{k}} \frac{\partial f}{\partial z} \right)}_{\text{gradient}} \cdot \hat{\mathbf{u}}$$



Summary - gradient in three coordinate systems

Cartesian:

$$\nabla f = \frac{\partial f}{\partial x} \hat{\mathbf{i}} + \frac{\partial f}{\partial y} \hat{\mathbf{j}} + \frac{\partial f}{\partial z} \hat{\mathbf{k}}$$

Circular cylindrical:

$$\nabla f = \frac{\partial f}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\boldsymbol{\theta}} + \frac{\partial f}{\partial z} \hat{\mathbf{k}}$$

Spherical:

$$\nabla f = \frac{\partial f}{\partial R} \hat{\mathbf{R}} + \frac{1}{R} \frac{\partial f}{\partial \phi} \hat{\boldsymbol{\phi}} + \frac{1}{R \sin(\phi)} \frac{\partial f}{\partial \theta} \hat{\boldsymbol{\theta}}$$

Studio classroom – Problem 4

Calculate the gradients of the given scalar fields expressed in terms of cylindrical or spherical coordinates:

(a) $f(r, \theta, z) = r\theta z$

(b) $f(R, \phi, \theta) = R\phi\theta$

Studio classroom – Problem 5

Let us revisit the static electric field $\mathbf{E}$ (conservative field), which can be determined from a known electric potential V using $\mathbf{E} = -\nabla V$

Calculate the electric field for the following two cases:

- (a) *In cylindrical coordinates*: an infinitely long line carrying a constant charge density ρ_L (C/m), for which the electric potential is given by
- $$V = -\frac{\rho_L}{2\pi\epsilon_0} \ln r.$$
- (b) *In spherical coordinates*: a point charge q (C) in the origin, for which the electric potential is given by $V = \frac{q}{4\pi\epsilon_0 R}$